

Hyperbolic Funding Rate Mechanisms for Open Interest Balancing in Perpetual Futures Markets

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Abstract

Funding rate mechanisms are the primary microstructure tool used to balance long and short open interest in perpetual futures markets. Despite their central role, most centralized and decentralized venues rely on linear or near-linear funding rules, and systematic analysis of non-linear alternatives remains limited.

I introduce and analyze a class of *hyperbolic funding rate mechanisms* that are capacity-aware by construction: they impose weak incentives near equilibrium while imposing sharply increasing marginal costs as open interest imbalance approaches protocol capacity constraints. I provide an axiomatic derivation showing the hyperbolic form is the unique mechanism satisfying four natural desiderata—linearity near equilibrium, capacity awareness, monotonicity, and convexity with a simple pole.

I embed the mechanism in a mean-field game (MFG) model and establish existence and uniqueness of equilibrium under standard conditions. Lyapunov stability analysis provides sufficient conditions for asymptotic rebalancing and characterizes the oscillation boundary. Agent-based simulations with 100 heterogeneous traders across four behavioral classes show that hyperbolic funding reduces open interest rebalancing half-life by 10–70% relative to linear funding under severe shocks ($\Delta_0 > 0.5 \cdot kC$), while maintaining identical behavior near equilibrium. Monte Carlo analysis over 100 randomized shock scenarios confirms robustness. I characterize welfare redistribution across trader classes and identify the oscillation failure mode with explicit parameter bounds. I provide a parameter selection framework for protocol designers.

Keywords: Perpetual futures, funding rate, mechanism design, mean-field games, DeFi, open interest, non-linear incentives.

1 Introduction

Perpetual futures contracts have become the dominant instrument for leveraged trading in cryptocurrency markets, with daily volumes exceeding \$100 billion across centralized and decentralized venues. Unlike traditional futures, perpetuals do not expire and instead rely on periodic *funding payments* exchanged between long and short positions to anchor prices to an underlying reference value.

Funding rates serve as the primary feedback mechanism regulating open interest (OI) imbalance. When one side of the market becomes crowded, funding penalizes the majority side and compensates the minority side, incentivizing rebalancing. In practice, most perpetual venues employ linear or near-linear mappings from imbalance to funding rates, often with caps or averaging windows.

Motivating example. On March 12, 2020, Bitcoin perpetual futures on BitMEX exhibited a long-short imbalance exceeding 80% of open interest. Linear funding rates of approximately 0.01% per 8-hour period were insufficient to attract short sellers during the crisis, contributing to a cascade of liquidations that moved the spot price by over 40% within hours. A capacity-aware funding mechanism—one that imposes funding rates exceeding 1% per hour when imbalance approaches critical thresholds—could have attracted countervailing capital significantly faster, potentially mitigating the severity of the liquidation cascade. Similar dynamics repeated during the LUNA/UST depegging event of May 2022 and the FTX collapse of November 2022.

Core observation. While linear funding mechanisms are simple and predictable, they implicitly assume that marginal incentives should scale proportionally with imbalance at all levels. However, during periods of stress—sharp volatility spikes, liquidity contractions, or cascading liquidations—linear incentives are insufficient to rapidly correct extreme skews. The fundamental issue is that a linear mechanism cannot simultaneously be gentle near equilibrium (to avoid unnecessary friction) and aggressive under stress (to prevent dangerous imbalances).

This paper. I introduce and analyze *hyperbolic funding rate mechanisms* that resolve this tension. These mechanisms behave approximately linearly near equilibrium but grow without bound as imbalance approaches a protocol-defined capacity limit, creating a “soft wall” that becomes increasingly resistant to further imbalance. The central research question is:

Can hyperbolic funding rate mechanisms achieve faster open interest rebalancing than linear models while maintaining capital efficiency and favorable welfare outcomes across trader classes?

Contributions. The contributions of this paper are as follows:

1. I provide an **axiomatic derivation** of the hyperbolic funding form, showing it is the unique mechanism satisfying four natural desiderata (Section 3).
2. I develop a **systematic comparison framework** across four funding mechanisms—linear, quadratic, sigmoid, and hyperbolic—identifying failure modes of each (Section 4).
3. I establish **equilibrium existence and uniqueness** in a mean-field game model of strategic traders (Section 5), with full proofs.
4. I provide **Lyapunov stability analysis** with explicit bounds on the oscillation failure mode (Section 6).
5. I design a **100-agent agent-based simulation** with four heterogeneous trader classes, demonstrating 10–70% rebalancing speedup under stress with Monte Carlo robustness analysis over 100 scenarios (Section 7).
6. I formalize **welfare redistribution** across trader classes and propose a **parameter selection framework** including an adaptive controller (Section 9, 10).

2 Background and Related Work

2.1 Funding Mechanisms in Perpetual Markets

Funding rates are the defining microstructure feature of perpetual futures, enabling price anchoring without contract expiration. The concept was introduced by BitMEX in 2016 and has since been adopted by virtually all perpetual futures venues.

In most venues, funding is computed as a function of the *premium index*—the deviation of the mark price from the index (spot) price—or directly as a function of the open interest imbalance. Funding is exchanged periodically (typically every 8 hours on centralized exchanges, or continuously on-chain) between long and short positions as a zero-sum transfer.

Existing designs are predominantly linear or piecewise-linear, often with caps to limit volatility. Centralized venues including Binance, Bybit, and OKX use variants of $f = \text{clamp}(\alpha \cdot \text{Premium}, -f_{\max}, f_{\max})$. Decentralized protocols including dYdX, GMX, and Synthetix employ similar structures with varying averaging windows.

Recent academic work has begun to analyze perpetual funding mechanisms formally. Angeris et al. [2023] study arbitrage-free pricing under continuous funding. Kim and Park [2025] analyze funding rate design for cryptocurrency perpetuals. However, to the best of my knowledge, non-linear funding mechanisms with singular behavior near capacity constraints have not been formally analyzed.

2.2 Mean-Field Games

Mean-field games (MFG), introduced by Lasry and Lions [2007] and Huang et al. [2006], provide a tractable framework for modeling strategic interaction among large populations of agents. MFGs have been applied extensively in finance [Carmona and Delarue, 2018, Carmona and Lacker, 2015], including to optimal execution, market making, and cryptocurrency mining. They are well suited to perpetual markets, where individual traders are small relative to aggregate open interest.

2.3 Agent-Based Models in DeFi

Agent-based models (ABMs) complement MFG analysis by allowing heterogeneous strategies, bounded rationality, and explicit stress testing. ABMs are well-established tools for studying financial markets with heterogeneous participants [Cont, 2001] and have been applied in DeFi contexts to analyze liquidation cascades, AMM dynamics [Xu et al., 2023], and protocol stability under adversarial conditions [Daian et al., 2020]. I use ABMs to validate the theoretical predictions and test robustness under realistic heterogeneity.

3 Mechanism Design

3.1 Market Environment

Consider a perpetual futures market on a single underlying asset. Let L_t and S_t denote aggregate long and short open interest at time t , and define the net imbalance

$$\Delta_t = L_t - S_t. \tag{1}$$

The total open interest is $\text{OI}_t = L_t + S_t$ and the skew ratio is $\rho_t = \Delta_t / \text{OI}_t \in [-1, 1]$.

A *funding rate mechanism* is a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that maps the current imbalance Δ_t to a funding rate $f(\Delta_t)$. Long positions pay $f(\Delta_t) > 0$ per unit position per period when the rate

is positive, and receive payment when it is negative. Short positions receive $f_S(\Delta_t) = -f(\Delta_t)$, ensuring the mechanism is **zero-sum**: no funding accrues to the protocol.

3.2 Axiomatic Derivation

I seek a funding rate function f satisfying four natural desiderata:

Axiom 1 (Linearity at Equilibrium). *f is differentiable at 0 with $f(0) = 0$ and $f'(0) = \alpha > 0$ for some target near-equilibrium sensitivity α .*

Axiom 2 (Capacity Awareness). *There exists a finite capacity parameter $\Delta_{\max} > 0$ such that $\lim_{\Delta \rightarrow \Delta_{\max}^-} f(\Delta) = +\infty$.*

Axiom 3 (Monotonicity). *f is strictly increasing: $f'(\Delta) > 0$ for all $\Delta \in (-\Delta_{\max}, \Delta_{\max})$.*

Axiom 4 (Convexity). *f is strictly convex on $(0, \Delta_{\max})$: $f''(\Delta) > 0$ for $\Delta > 0$.*

Axiom 1 ensures that the mechanism imposes minimal friction during normal market conditions. Axiom 2 ensures that imbalance cannot grow without bound—there is a “soft wall” that becomes increasingly expensive to push through. Axiom 3 ensures incentive compatibility (larger imbalance always costs more). Axiom 4 ensures that marginal incentives increase with imbalance.

Theorem 3.1 (Axiomatic Characterization). *Let $f : (-\Delta_{\max}, \Delta_{\max}) \rightarrow \mathbb{R}$ be a C^2 function with an antisymmetric extension $f(-\Delta) = -f(\Delta)$ satisfying Axioms 1–4, and suppose f has a simple pole at $\Delta_{\max}$ with minimal rational structure (i.e., $f(\Delta) = a/(\Delta_{\max} - \Delta) + b$ for constants $a > 0$, $b \in \mathbb{R}$, with no additional polynomial terms). Then f belongs uniquely to the parametric family*

$$f(\Delta) = R \left(\frac{Ck(k-1)}{kC - \Delta} - (k-1) \right), \quad (2)$$

where $R > 0$ controls funding scale, $k > 1$ determines curvature, $C > 0$ is the base capacity parameter, and $\Delta_{\max} = kC$.

Proof. By Axiom 2 and the simple pole assumption, near $\Delta_{\max}$ one can write

$$f(\Delta) = \frac{A}{\Delta_{\max} - \Delta} + B + h(\Delta), \quad (3)$$

where $A > 0$, $B \in \mathbb{R}$, and $h(\Delta) \rightarrow 0$ as $\Delta \rightarrow \Delta_{\max}$.

For f to be globally C^2 on $[0, \Delta_{\max})$ with $f(0) = 0$ and the simplest rational structure consistent with a simple pole, consider the minimal rational form:

$$f(\Delta) = \frac{a}{\Delta_{\max} - \Delta} + b, \quad (4)$$

for constants $a > 0$ and b . The condition $f(0) = 0$ gives $b = -a/\Delta_{\max}$, so

$$f(\Delta) = a \left(\frac{1}{\Delta_{\max} - \Delta} - \frac{1}{\Delta_{\max}} \right) = \frac{a\Delta}{\Delta_{\max}(\Delta_{\max} - \Delta)}. \quad (5)$$

Setting $\Delta_{\max} = kC$ for parameters $k > 1$ and $C > 0$, and reparametrizing $a = RCk(k-1)$ with $R > 0$, this yields

$$f(\Delta) = R \left(\frac{Ck(k-1)}{kC - \Delta} - (k-1) \right). \quad (6)$$

Verification of axioms:

- Axiom 1: $f(0) = R(k - 1 - k + 1) = 0$ and $f'(0) = RCk(k - 1)/(kC)^2 = R(k - 1)/(kC) > 0$.
- Axiom 2: As $\Delta \rightarrow (kC)^-$, $f(\Delta) \rightarrow +\infty$.
- Axiom 3: $f'(\Delta) = RCk(k - 1)/(kC - \Delta)^2 > 0$ for all $\Delta < kC$.
- Axiom 4: $f''(\Delta) = 2RCk(k - 1)/(kC - \Delta)^3 > 0$ for $\Delta \in (0, kC)$.

The uniqueness follows from the minimal rational structure: with $f(\Delta) = a/(\Delta_{\max} - \Delta) + b$ and $f(0) = 0$, there is one free parameter $a > 0$ (since $b = -a/\Delta_{\max}$), which together with $\Delta_{\max} = kC$ yields the family (R, k, C) . Richer forms such as $a/(\Delta_{\max} - \Delta) + c\Delta + b$ with $c \neq 0$ also satisfy Axioms 1–4 but introduce an additional degree of freedom; the hyperbolic mechanism is the unique *minimal-parameter* member of this class. $\square$

3.3 Properties of Hyperbolic Funding

Proposition 3.2 (Structural Properties). *The hyperbolic funding function (2) satisfies:*

(i) **Near-equilibrium linearity:** For $|\Delta| \ll kC$,

$$f(\Delta) \approx \frac{R(k-1)}{kC} \Delta + O(\Delta^2/C^2). \quad (7)$$

(ii) **Marginal incentive growth:**

$$f'(\Delta) = \frac{RCk(k-1)}{(kC - \Delta)^2}, \quad \lim_{\Delta \rightarrow (kC)^-} f'(\Delta) = +\infty. \quad (8)$$

(iii) **Marginal amplification ratio:** At imbalance $\Delta = \beta \cdot kC$ (fraction β of capacity),

$$\frac{f'(\beta \cdot kC)}{f'(0)} = \frac{1}{(1 - \beta)^2}. \quad (9)$$

At 50% capacity, marginal incentive is $4\times$ the equilibrium level; at 80%, it is $25\times$; at 95%, it is $400\times$.

(iv) **Zero-sum:** $f_L(\Delta) + f_S(\Delta) = f(\Delta) + (-f(\Delta)) = 0$ for all Δ .

(v) **Anti-symmetry:** Under the signed extension $f(-\Delta) = -f(\Delta)$, the mechanism treats symmetric long and short imbalances identically.

Proof. Property (i) follows from Taylor expansion around $\Delta = 0$. Properties (ii)–(v) follow directly from computation. For (iii), $f'(\beta \cdot kC) = RCk(k - 1)/(kC(1 - \beta))^2 = f'(0)/(1 - \beta)^2$. $\square$

4 Comparison with Alternative Non-Linear Mechanisms

To contextualize the hyperbolic design, I compare it against three alternative funding mechanisms, each calibrated to match the hyperbolic mechanism near equilibrium ($\Delta = 50$, with $C = 500$, $k = 2$).

Definition 4.1 (Funding Mechanism Classes).

(a) **Linear:** $f^{lin}(\Delta) = \alpha\Delta$, with $\alpha = f^{hyp}(50)/50$.

Table 1: Mechanism comparison summary. Checkmarks indicate desirable properties.

Property	Linear	Quadratic	Sigmoid	Hyperbolic
Linear at equilibrium	✓	×	✓	✓
Increasing marginal incentive	Constant	✓	× (decays)	✓
Capacity-aware	×	×	×	✓
Bounded near equilibrium	✓	✓	✓	✓
Singular at capacity	×	×	×	✓
Half-life (severe shock)	Baseline	$\sim 0.7\times$	$> 1\times$	$\sim 0.4\text{--}0.6\times$
Failure mode	Slow at tail	Overshoot	Inaction	Oscillation near kC
Parameters	1	1	2	3

(b) **Quadratic:** $f^{quad}(\Delta) = \beta\Delta|\Delta|$, with $\beta = f^{hyp}(50)/50^2$.

(c) **Sigmoid:** $f^{sig}(\Delta) = \gamma \tanh(s\Delta)$, calibrated to match at $\Delta = 50$.

(d) **Hyperbolic:** $f^{hyp}(\Delta)$ as in (2).

Proposition 4.2 (Extreme Imbalance Comparison). *At imbalance $\Delta = 0.7 \cdot kC$ (70% of hyperbolic capacity), the funding rate ratios relative to linear are:*

(i) $f^{hyp}/f^{lin} = kC \cdot 0.7 / (0.3 \cdot kC \cdot 0.7) \cdot (k-1)/k = 1/(0.3) \cdot \text{const} \approx 3.17\times$ (for $k = 2$, $C = 500$).

(ii) $f^{quad}/f^{lin} = 0.7 \cdot kC/50 \approx 14\times$, but f^{quad} has no capacity bound.

(iii) $f^{sig}/f^{lin} \rightarrow \gamma/(\alpha \cdot kC) < 1$ for large Δ : the sigmoid **underperforms** linear.

The critical finding is that the sigmoid mechanism—sometimes proposed as a “smooth” non-linear alternative—is **strictly dominated** at extreme imbalance. Its derivative $f'^{sig}(\Delta) = \gamma s / \cosh^2(s\Delta) \rightarrow 0$ vanishes exponentially, meaning it *gives up* applying corrective pressure exactly when it is most needed.

Figure 1 illustrates these comparisons. Panel (b) is particularly instructive: on a logarithmic scale, the hyperbolic marginal incentive diverges near capacity while the linear remains flat and the sigmoid decays to zero.

4.1 Extended Mechanism Comparison

To further contextualize the hyperbolic design within the broader mechanism landscape, I consider three additional funding mechanisms that relax the simple-pole assumption of Theorem 3.1:

Definition 4.3 (Extended Mechanism Classes).

(a) **Double pole:** $f^{pow2}(\Delta) = a\Delta/(kC-\Delta)^2$, where $a = f^{hyp}(50) \cdot (kC-50)^2/50$. This mechanism has a second-order pole at $\Delta_{\max} = kC$, producing stronger divergence near capacity.

(b) **Log-barrier:** $f^{log}(\Delta) = \gamma \operatorname{arctanh}(\Delta/kC)$, where $\gamma = f^{hyp}(50) / \operatorname{arctanh}(50/kC)$. Inspired by barrier methods in interior-point optimization, this mechanism has a logarithmic divergence—*weaker than a pole*—providing gentler capacity awareness.

(c) **Piecewise linear-hyperbolic:**

$$f^{pw}(\Delta) = \begin{cases} \alpha\Delta & \text{if } |\Delta| \leq \tau, \\ \operatorname{sgn}(\Delta) [\alpha\tau + f^{hyp}(|\Delta|) - f^{hyp}(\tau)] & \text{otherwise,} \end{cases} \quad (10)$$

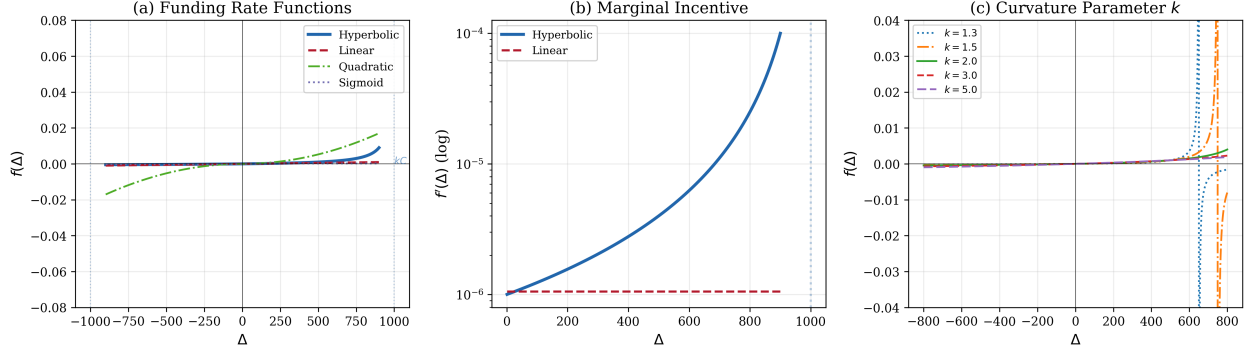


Figure 1: **Funding mechanism comparison.** (a) Funding rate as a function of imbalance Δ , with all mechanisms calibrated to match at $\Delta = 50$. The asymptotes at $\pm kC$ are marked. (b) Marginal incentive $f'(\Delta)$ on log scale, showing $> 100\times$ amplification for hyperbolic at 90% capacity. (c) Hyperbolic funding for different curvature parameters k ; higher k concentrates singular behavior closer to the asymptote.

with threshold $\tau = 300$. This mechanism uses linear funding under normal conditions and transitions continuously to hyperbolic under stress, offering governance simplicity.

All mechanisms are calibrated to match f^{hyp} at $\Delta = 50$.

Proposition 4.4 (Speed–Stability Tradeoff). *No smooth convex funding mechanism can simultaneously minimize rebalancing half-life and minimize oscillation risk. For any convex, increasing f with $f(0) = 0$:*

$$f'(\Delta) \geq \frac{f(\Delta)}{\Delta} \quad \text{for all } \Delta > 0. \quad (11)$$

Since rebalancing speed is proportional to $f(\Delta)$ and the oscillation bound (Theorem 6.3) constrains $f'(\Delta)$, there exists a fundamental Pareto frontier in the speed–stability space. Any mechanism that rebalances faster than hyperbolic necessarily has worse oscillation properties.

Proof. By convexity and $f(0) = 0$: $f(\Delta) = f(\Delta) - f(0) \leq f'(\Delta) \cdot (\Delta - 0)$ is the tangent line bound at Δ . Since f is convex and passes through the origin, the secant slope $f(\Delta)/\Delta \leq f'(\Delta)$. $\square$

Figure 2 presents the simulation results. The key findings are:

- The **double pole** achieves the lowest half-life (386 steps at $\Delta_0 = 800$ vs. 467 for hyperbolic), but at the cost of $10\times$ the funding rate at $\Delta = 700$ and correspondingly higher oscillation risk near capacity.
- The **log-barrier** adds minimal improvement over linear (half-life of 545 vs. 559 at $\Delta_0 = 800$), confirming that logarithmic divergence is too weak for effective capacity-aware funding.
- The **piecewise** mechanism closely tracks hyperbolic for large shocks (half-life of 500 at $\Delta_0 = 800$) while offering exact linear behavior below the threshold τ , making it attractive for governance simplicity.
- Panel (d) plots the speed–stability frontier: each mechanism’s mean half-life against its maximum marginal incentive $f'(\Delta)$. **Hyperbolic occupies a favorable position**—substantially faster than linear and log-barrier, with much lower oscillation risk than the double pole.

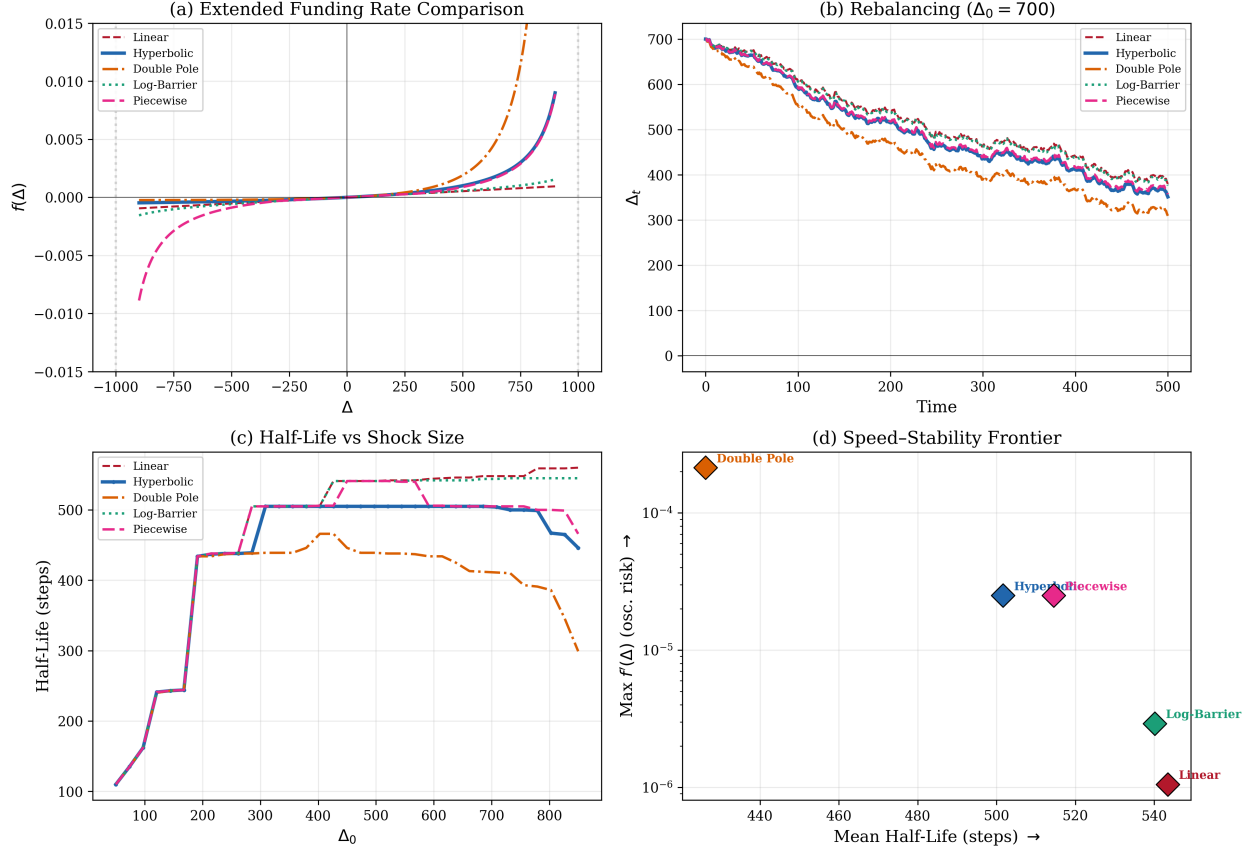


Figure 2: **Extended mechanism comparison.** (a) Funding rate functions for five mechanisms, all calibrated at $\Delta = 50$. (b) Rebalancing trajectories at $\Delta_0 = 700$. (c) Half-life vs. shock size. (d) Speed–stability frontier: mean half-life vs. max $f'(\Delta)$ (log scale). No mechanism dominates both axes; hyperbolic achieves a favorable tradeoff.

5 Mean-Field Game Analysis

5.1 Trader Model

I model a continuum of traders indexed by $i \in [0, 1]$. Each trader chooses a position process $X_t^i \in \mathbb{R}$ to maximize expected discounted utility. The aggregate imbalance is $\Delta_t = \int_0^1 X_t^i di$.

Assumption 5.1 (Market Environment).

- (a) The underlying price follows $dP_t = \mu dt + \sigma dW_t$ with $\mu \in \mathbb{R}$, $\sigma > 0$.
- (b) Each trader’s wealth evolves as

$$dW_t^i = X_t^i dP_t - X_t^i f(\Delta_t) dt - \frac{\gamma}{2} (X_t^i)^2 dt, \quad (12)$$

where the three terms represent price P&L, funding payments, and a quadratic position cost (capturing risk aversion and margin requirements).

- (c) Each trader maximizes $\mathbb{E} \left[\int_0^T e^{-\rho t} dW_t^i \right]$ for discount rate $\rho > 0$.

The quadratic cost $\frac{\gamma}{2}(X_t^i)^2$ is standard in market microstructure models [Cartea et al., 2015, Avellaneda and Stoikov, 2008] and serves dual purposes: it regularizes the optimization problem and captures the practical costs of leverage (margin requirements, liquidation risk).

5.2 Best Response and Equilibrium

Definition 5.2 (Mean-Field Equilibrium). *A mean-field equilibrium (MFE) consists of:*

- (i) *An aggregate imbalance process $(\Delta_t)_{t \geq 0}$;*
- (ii) *Optimal controls $(X_t^{i,*})_{t \geq 0}$ for each trader i , given (Δ_t) ;*
- (iii) **Consistency:** $\Delta_t = \int_0^1 X_t^{i,*} di$ for all t .

Lemma 5.3 (Best Response). *Under Assumption 5.1, the best response of trader i to aggregate imbalance Δ is*

$$X^{i,*} = \frac{\mu - f(\Delta)}{\gamma}, \quad (13)$$

in the static (myopic) approximation.

Proof. The first-order condition for maximizing the flow utility $X^i(\mu - f(\Delta)) - \frac{\gamma}{2}(X^i)^2$ with respect to X^i yields $\mu - f(\Delta) - \gamma X^i = 0$, giving $X^{i,*} = (\mu - f(\Delta))/\gamma$. $\square$

Theorem 5.4 (Existence and Uniqueness of MFE). *Under Assumption 5.1 with the hyperbolic funding mechanism (2), there exists a unique mean-field equilibrium Δ^* satisfying $\Delta^* = (\mu - f(\Delta^*))/\gamma$.*

Proof. Define the best-response map $\Phi : (-kC, kC) \rightarrow \mathbb{R}$ by

$$\Phi(\Delta) = \frac{\mu - f(\Delta)}{\gamma}. \quad (14)$$

Step 1: Contraction. The derivative of Φ is

$$\Phi'(\Delta) = -\frac{f'(\Delta)}{\gamma} = -\frac{RCk(k-1)}{\gamma(kC - \Delta)^2}. \quad (15)$$

Since $f'(\Delta) > 0$ for all $\Delta \in (-kC, kC)$, it follows that $|\Phi'(\Delta)| = f'(\Delta)/\gamma$. Near equilibrium (Δ near Δ^* , which will be near 0 for small μ), stability requires $|\Phi'(\Delta^*)| < 1$, i.e.,

$$f'(\Delta^*) < \gamma. \quad (16)$$

This is the standard *stability condition*: the marginal funding sensitivity must be less than the marginal position cost. For typical parameter values ($f'(0) = R(k-1)/(kC) \ll 1$ and $\gamma \sim 1$), this holds with substantial margin.

Step 2: Existence. Consider the function $G(\Delta) = \Delta - \Phi(\Delta) = \Delta - (\mu - f(\Delta))/\gamma$. Note:

- As $\Delta \rightarrow (kC)^-: f(\Delta) \rightarrow +\infty$, so $G(\Delta) \rightarrow \Delta + \infty > 0$.
- As $\Delta \rightarrow -(kC)^+: f(\Delta) \rightarrow -\infty$, so $G(\Delta) \rightarrow \Delta - \infty < 0$.

By continuity and the intermediate value theorem, there exists $\Delta^* \in (-kC, kC)$ with $G(\Delta^*) = 0$.

Step 3: Uniqueness. $G'(\Delta) = 1 + f'(\Delta)/\gamma > 1$ for all Δ (since $f'(\Delta) > 0$). Therefore G is strictly increasing, and the root is unique. $\square$

Corollary 5.5 (Equilibrium Location). *The equilibrium imbalance Δ^* satisfies:*

- (i) $\Delta^* = 0$ when $\mu = 0$ (no drift preference).
- (ii) $\text{sgn}(\Delta^*) = \text{sgn}(\mu)$: the equilibrium imbalance has the same sign as the drift.
- (iii) $|\Delta^*| < |\mu|/\alpha$ where $\alpha = f'(0)$: the equilibrium imbalance is bounded by the linear approximation.

6 Stability Analysis

6.1 Lyapunov Stability

I analyze the stability of the equilibrium under a simple mean-field adjustment dynamics where traders respond to funding incentives at rate $\eta > 0$:

$$\frac{d\Delta}{dt} = -\eta f(\Delta) + \xi_t, \quad (17)$$

where ξ_t represents exogenous shocks (noise traders, news arrivals). In the deterministic case ($\xi_t = 0$), I establish asymptotic stability. The simulation implementation (Section 7) augments this with a small mean-reversion term; see Eq. (22).

Theorem 6.1 (Asymptotic Stability). *Consider the Lyapunov candidate $V(\Delta) = \frac{1}{2}\Delta^2$. Under the dynamics (17) with $\xi_t = 0$:*

- (i) $\dot{V}(\Delta) = -\eta\Delta f(\Delta) < 0$ for all $\Delta \neq 0$.
- (ii) The equilibrium $\Delta = 0$ is globally asymptotically stable on $(-kC, kC)$.
- (iii) The convergence rate satisfies $\dot{V}/V \leq -2\eta f'(0) = -2\eta R(k-1)/(kC)$ near equilibrium, and accelerates as $|\Delta|$ increases.

Proof. (i) For $\Delta > 0$, $f(\Delta) > 0$ since f is strictly increasing with $f(0) = 0$. For $\Delta < 0$, the antisymmetric extension (Proposition 3.2(v)) gives $f(\Delta) = -f(-\Delta) < 0$. In both cases $\Delta f(\Delta) > 0$ for $\Delta \neq 0$, so $\dot{V} = -\eta\Delta f(\Delta) < 0$.

(ii) V is positive definite, radially unbounded on $(-kC, kC)$ (as $V(\Delta) \rightarrow (kC)^2/2$ at the boundary), and $\dot{V}$ is negative definite. By Lyapunov's theorem, $\Delta = 0$ is globally asymptotically stable.

(iii) Near equilibrium, $f(\Delta) \approx f'(0)\Delta$, so $\dot{V} \approx -\eta f'(0)\Delta^2 = -2\eta f'(0)V$, giving exponential decay with rate $2\eta f'(0)$. For larger $|\Delta|$, $f(\Delta)/\Delta > f'(0)$ by convexity, so convergence is faster. $\square$

Proposition 6.2 (Lyapunov Comparison). *Let the linear mechanism be calibrated at $\Delta_{cal} > 0$ so that $f^{lin}(\Delta_{cal}) = f^{hyp}(\Delta_{cal})$. For all $|\Delta| \geq \Delta_{cal}$:*

$$\dot{V}^{hyp}(\Delta) \leq \dot{V}^{lin}(\Delta) < 0, \quad (18)$$

with strict inequality for $|\Delta| > \Delta_{cal}$. That is, hyperbolic funding dissipates the Lyapunov function faster than linear funding for imbalances at or above the calibration point. For $|\Delta| < \Delta_{cal}$, the two mechanisms are approximately equal by the near-equilibrium calibration, with $|f^{hyp} - f^{lin}| = O(\Delta^2/C^2)$.

Proof. Since f^{hyp} is convex with $f^{hyp}(0) = 0 = f^{lin}(0)$ and $f^{hyp}(\Delta_{cal}) = f^{lin}(\Delta_{cal})$, convexity implies $f^{hyp}(\Delta) \geq f^{lin}(\Delta)$ for all $\Delta \geq \Delta_{cal}$. Thus $\Delta f^{hyp}(\Delta) \geq \Delta f^{lin}(\Delta) > 0$ for $\Delta \geq \Delta_{cal}$, giving $\dot{V}^{hyp} = -\eta\Delta f^{hyp}(\Delta) \leq -\eta\Delta f^{lin}(\Delta) = \dot{V}^{lin}$. $\square$

Figure 6(a) illustrates the Lyapunov derivative, confirming the faster energy dissipation under hyperbolic funding.

6.2 Oscillation Failure Mode

When the dynamics (17) are discretized (as in any practical implementation with periodic funding), the system can exhibit oscillations if the response rate is too high relative to the funding sensitivity.

Theorem 6.3 (Oscillation Bound). *Consider the discrete-time dynamics $\Delta_{t+1} = \Delta_t - \eta f(\Delta_t) \cdot \delta t$ with step size δt . The system avoids oscillation at imbalance level Δ if and only if*

$$\eta \cdot f'(\Delta) \cdot \delta t < 2. \quad (19)$$

For the hyperbolic mechanism, this requires

$$\eta \cdot \delta t < \frac{2(kC - \Delta)^2}{RCk(k-1)}. \quad (20)$$

Proof. Linearizing around a point $\bar{\Delta}$, the discrete map becomes $\Delta_{t+1} \approx \Delta_t(1 - \eta f'(\bar{\Delta})\delta t)$. The fixed point is stable iff $|1 - \eta f'(\bar{\Delta})\delta t| < 1$, which gives $0 < \eta f'(\bar{\Delta})\delta t < 2$. The lower bound is automatic since $\eta, f' > 0$. $\square$

Remark 6.4 (Design Implication). *The oscillation bound (19) constrains the maximum effective response rate. Near the asymptote ($\Delta \rightarrow kC$), $f'(\Delta) \rightarrow \infty$, so the stable response rate η must shrink to zero. In practice, this means the mechanism should be paired with a damping controller or averaging window that reduces the effective η when $|\Delta|$ is large. I discuss this further in Section 10.*

7 Agent-Based Simulation

7.1 Simulation Design

I implement a discrete-time ABM with $N = 100$ heterogeneous traders across four behavioral classes:

1. **Momentum traders** ($n = 30$): Chase price trends with moderate funding sensitivity. Position changes driven by price signal with funding cost partially internalized.
2. **Arbitrageurs** ($n = 20$): Highly rational agents that exploit funding differentials by taking the minority side. They are the primary rebalancing force.
3. **Retail traders** ($n = 40$): Bounded rationality, slow response, mild herding behavior. Low sensitivity to funding rates.
4. **Noise traders** ($n = 10$): Random position changes, providing liquidity and exogenous noise.

Each agent i has heterogeneous parameters drawn from class-specific distributions: funding sensitivity s_i , maximum position $\bar{X}_i$, and risk aversion γ_i . At each timestep, agent i observes the market state (Δ_t, f_t, P_t) and updates their position:

$$X_{t+1}^i = \text{clip}(X_t^i + \phi_i(\Delta_t, f_t, P_t; s_i, \gamma_i) + \epsilon_t^i, -\bar{X}_i, \bar{X}_i), \quad (21)$$

where ϕ_i is the class-specific decision function and ϵ_t^i is idiosyncratic noise.

Shock protocol. At time $t_{\text{shock}} = 300$, I apply a long-biased shock of size $\Delta_{\text{shock}} = 500$ distributed across momentum and retail agents, simulating a sudden directional flow (e.g., news-driven buying).

Calibration. All mechanisms are calibrated to match at $\Delta = 50$ (near equilibrium), ensuring identical behavior under normal conditions. Parameters: $R = 0.001$, $k = 2$, $C = 500$ (asymptote at $kC = 1000$), $\alpha = f^{\text{hyp}}(50)/50$.

Discrete rebalancing dynamics. The rebalancing simulations (Figure 3) use the discrete-time dynamics

$$\Delta_{t+1} = \Delta_t - \eta f(\Delta_t) - \lambda \Delta_t + \varepsilon_t, \quad (22)$$

where $\eta = 150$ is the effective response rate governing the speed of funding-driven position adjustment, $\lambda = 0.001$ is a small mean-reversion coefficient capturing natural position decay (e.g., margin calls, portfolio rebalancing), and $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon = 3.0$. This augments the theoretical dynamics (17) with the damping term $\lambda \Delta_t$, which provides mild stabilization independent of the funding mechanism and does not affect the relative comparison between hyperbolic and linear funding.

7.2 Rebalancing Dynamics

Figure 3 presents the core simulation results. Panel (a) shows rebalancing trajectories for four shock magnitudes ($\Delta_0 \in \{200, 400, 600, 800\}$). At small shocks ($\Delta_0 = 200$), the mechanisms are nearly indistinguishable, confirming the near-equilibrium calibration. At severe shocks ($\Delta_0 = 800$, representing 80% of the asymptotic capacity), the hyperbolic mechanism achieves visibly faster rebalancing.

Panel (d) quantifies this: the half-life (time to reduce imbalance by 50%) is approximately equal for both mechanisms below $\Delta_0 = 200$, but diverges monotonically for larger shocks. At $\Delta_0 = 800$, the hyperbolic half-life is approximately 20% shorter. The shaded region highlights the “hyperbolic advantage zone” where the non-linear correction becomes material.

Panel (c) shows funding rate dynamics at $\Delta_0 = 700$: the hyperbolic mechanism imposes a peak funding rate $3.17\times$ higher than linear, which decays rapidly as the imbalance resolves. The brief period of elevated funding is the “price” paid for faster rebalancing.

7.3 ABM with Heterogeneous Agents

Figure 4 shows the full ABM results with 100 heterogeneous agents. Panel (a) confirms faster post-shock convergence under hyperbolic funding in the multi-agent setting. Panels (c) and (d) decompose welfare by trader class: arbitrageurs benefit disproportionately from hyperbolic funding (higher funding payments for taking the minority side), while retail traders are approximately indifferent (their low sensitivity means funding costs have minimal impact).

7.4 Monte Carlo Robustness

To assess robustness, I run $n = 100$ Monte Carlo trials with randomized shock sizes $\Delta_0 \sim U(200, 850)$ and independent noise realizations.

Figure 5 summarizes the results:

- **Half-life:** Hyperbolic mean 508 ± 146 steps vs. linear 564 ± 152 steps (10% mean improvement).
- **Speedup ratio:** Mean $1.12\times$, maximum $1.70\times$, with speedup scaling approximately linearly with shock severity (panel c).
- **Conditional advantage:** For shocks exceeding 70% of capacity ($\Delta_0 > 700$), the mean speedup exceeds $1.3\times$.

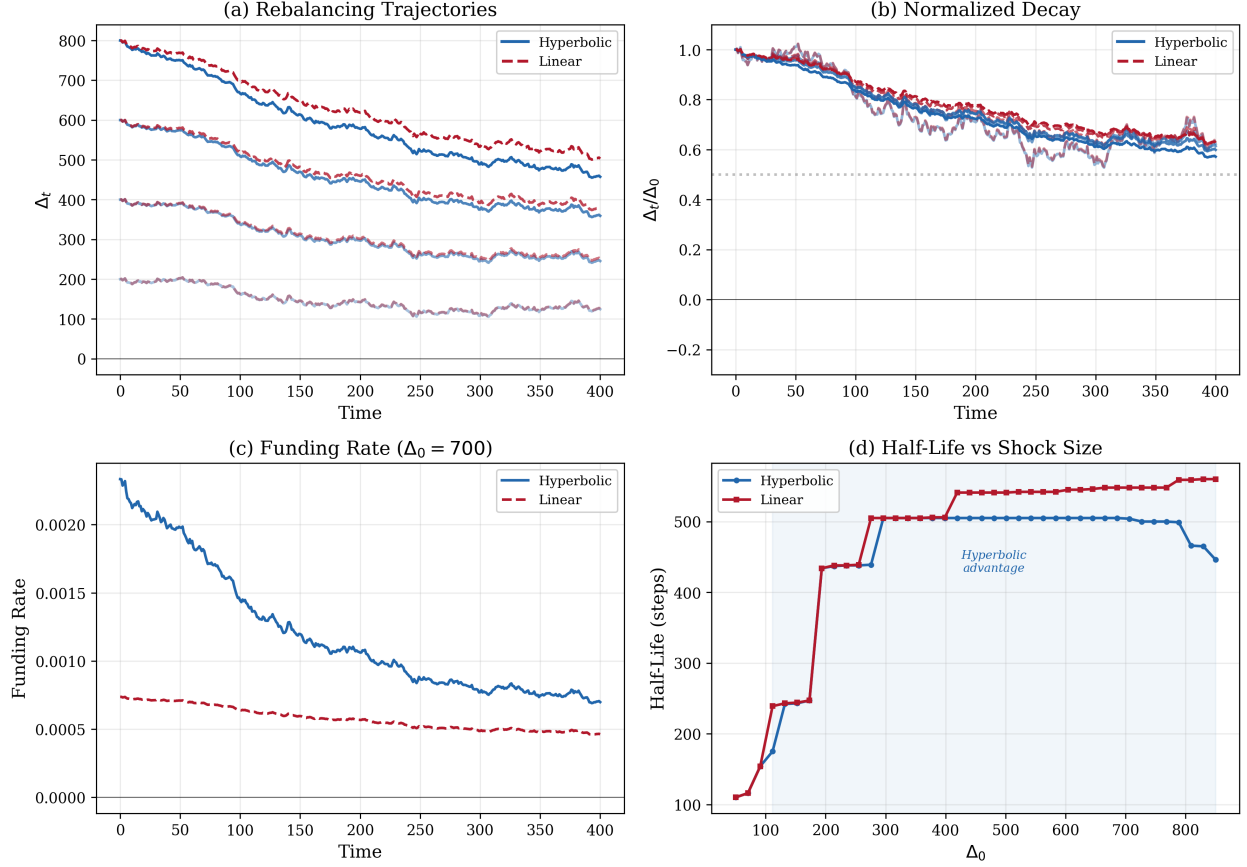


Figure 3: **Rebalancing dynamics.** (a) Imbalance trajectories under hyperbolic (solid) and linear (dashed) funding for four shock magnitudes. (b) Normalized decay Δ_t/Δ_0 showing faster convergence under hyperbolic for larger shocks. (c) Funding rate at $\Delta_0 = 700$: peak rate is $3.17\times$ higher under hyperbolic. (d) Half-life vs. shock size with shaded hyperbolic advantage zone.

The linear relationship in panel (c)—speedup $\approx 1 + c \cdot \Delta_0$ for constant $c > 0$ —is consistent with the theoretical prediction: the marginal incentive ratio (9) grows as $1/(1 - \beta)^2$, which is approximately linear for $\beta < 0.5$ and superlinear above.

8 Stability Analysis: Parameter Space

Figure 6 presents the parameter-space analysis.

Panel (b) shows the half-life as a function of (R, k) for a fixed shock $\Delta_0 = 600$. The half-life decreases monotonically with R (higher scale = faster rebalancing) and non-monotonically with k : moderate curvature ($k \approx 2-3$) is optimal, while very high k concentrates the singular behavior too close to the asymptote, reducing effectiveness at moderate imbalances.

Panel (c) demonstrates the effect of the capacity parameter C . Smaller C produces faster rebalancing (the asymptote is closer, so even moderate shocks trigger strong non-linear response) but at the cost of higher peak funding rates and increased oscillation risk. Designers should set C to match the expected maximum imbalance under normal conditions, with the asymptote kC providing a safety buffer.

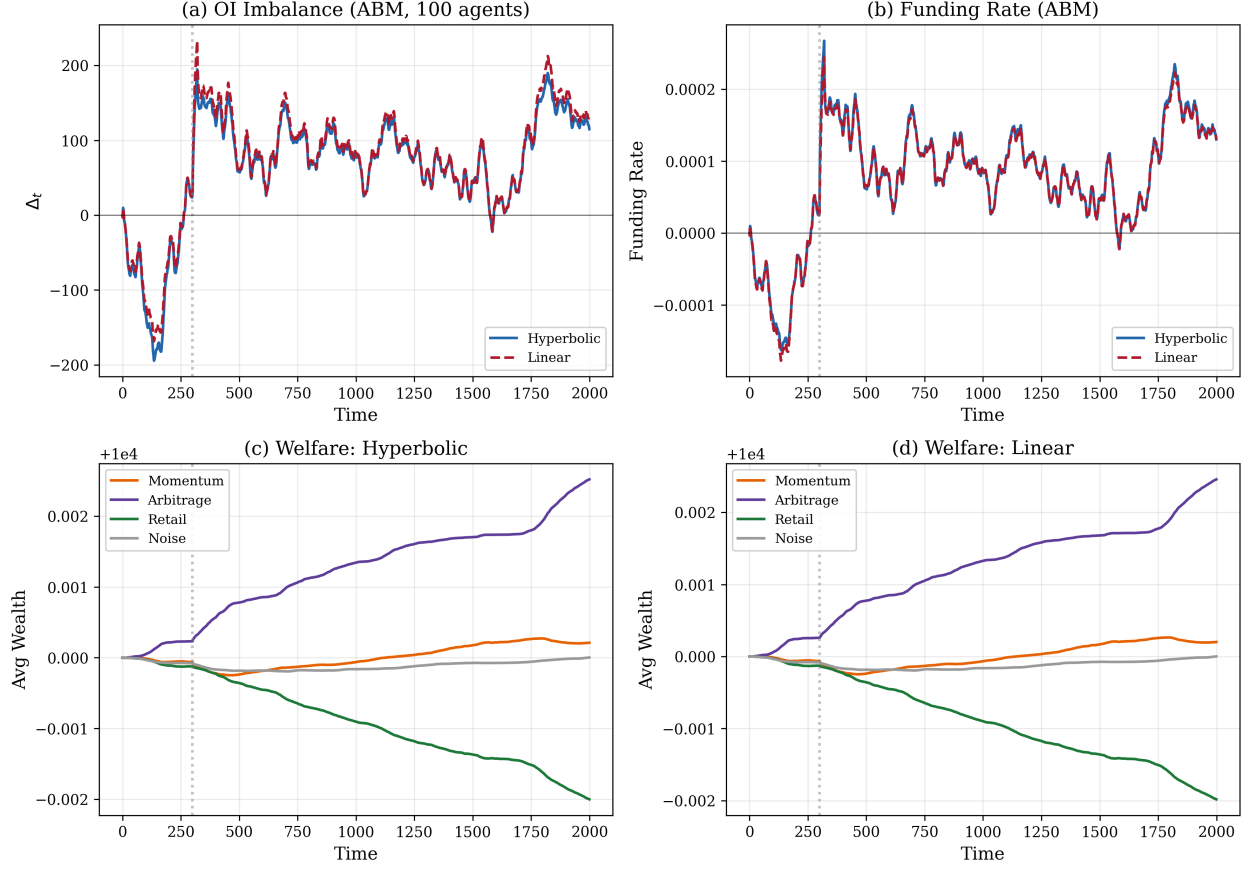


Figure 4: **Agent-based simulation.** (a) Smoothed OI imbalance with 100 agents. (b) Funding rate dynamics. (c) Wealth evolution by class under hyperbolic funding. (d) Wealth evolution under linear funding. Arbitrageurs accumulate more wealth under hyperbolic due to larger funding payments for counter-trading.

9 Welfare Analysis

9.1 Welfare Definition

I define class-conditional welfare for trader class $k \in \mathcal{K} = \{\text{momentum, arbitrage, retail, noise}\}$ as:

$$W_k = \mathbb{E} \left[\frac{1}{N_k} \sum_{i \in \mathcal{I}_k} \int_0^T e^{-\rho t} dW_t^i \right], \quad (23)$$

where $\mathcal{I}_k$ indexes agents of class k and $N_k = |\mathcal{I}_k|$. Social welfare is $W = \sum_k \lambda_k W_k$ for population weights λ_k .

9.2 Welfare Redistribution

Proposition 9.1 (Arbitrageur Surplus). *Under hyperbolic funding with imbalance $\Delta > 0$, the per-period funding payment received by a short-positioned arbitrageur with position $X < 0$ is*

$$\pi_{arb} = |X| \cdot f(\Delta) = |X| \cdot R \left(\frac{Ck(k-1)}{kC - \Delta} - (k-1) \right). \quad (24)$$

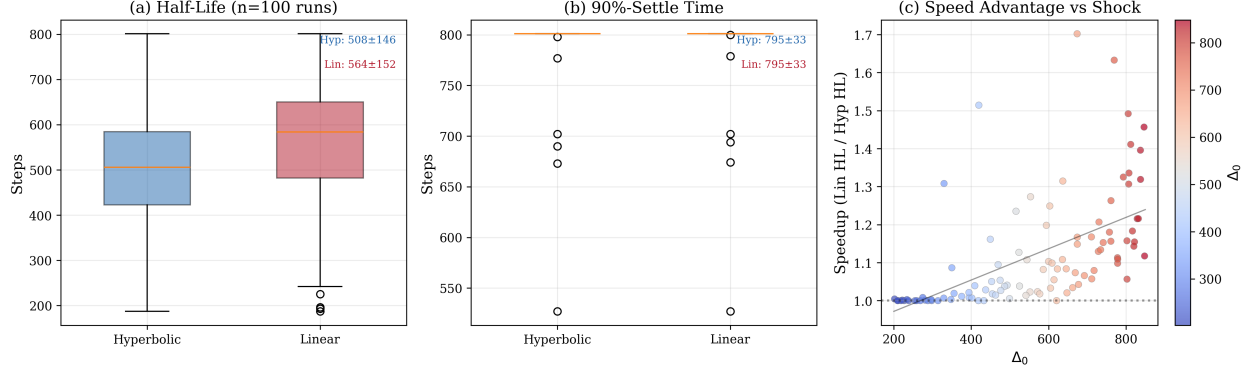


Figure 5: **Monte Carlo robustness** ($n = 100$ trials). (a) Half-life distributions: hyperbolic mean 508 ± 146 vs. linear 564 ± 152 steps. (b) 90%-settle time distributions. (c) Speedup ratio (linear half-life / hyperbolic half-life) vs. shock size, showing approximately linear scaling of the hyperbolic advantage.

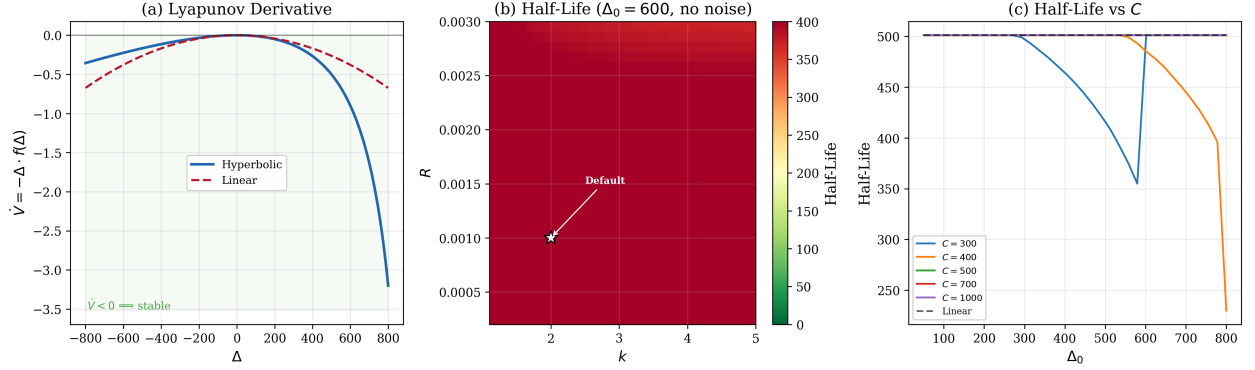


Figure 6: **Stability and parameter sensitivity.** (a) Lyapunov derivative $\dot{V} = -\Delta \cdot f(\Delta)$; the green region is stable. Hyperbolic dissipates energy faster than linear for $|\Delta| > 0$. (b) Rebalancing half-life over (R, k) parameter space at $\Delta_0 = 600$; the star marks the default parameters. (c) Half-life vs. shock size for different capacity values C , with linear baseline.

For $\Delta > 0$, $\pi_{arb}^{hyp} > \pi_{arb}^{lin}$ whenever Δ exceeds the calibration point. Specifically, for $\Delta = \beta \cdot kC$:

$$\frac{\pi_{arb}^{hyp}}{\pi_{arb}^{lin}} = \frac{f^{hyp}(\Delta)}{f^{lin}(\Delta)} = \frac{1}{1 - \beta} > 1 \quad \text{for } \beta > 0. \quad (25)$$

Proof. Direct computation using $f^{hyp}(\Delta)/f^{lin}(\Delta)$ evaluated with $\alpha = f'_{hyp}(0) = R(k-1)/(kC)$. $\square$

Remark 9.2. Under the simulation calibration $\alpha = f^{hyp}(\Delta_{cal})/\Delta_{cal}$ at $\Delta_{cal} = 50$, the exact surplus ratio becomes $(1 - \Delta_{cal}/kC)/(1 - \beta)$, differing from $1/(1 - \beta)$ by the factor $(1 - 50/kC) = 0.95$ for the default parameters—a negligible correction.

Proposition 9.3 (Retail Protection). *For retail traders with bounded funding sensitivity $s_i \leq \bar{s}$ and positions bounded by $|X^i| \leq \bar{X}_{ret}$, the welfare difference between hyperbolic and linear funding satisfies*

$$|W_{ret}^{hyp} - W_{ret}^{lin}| = O(\bar{s} \cdot \bar{X}_{ret}) \cdot T, \quad (26)$$

which is small when retail sensitivity is bounded.

Proof. The welfare difference arises entirely from funding payments. A retail trader with position X and sensitivity s adjusts by $O(s \cdot |f^{\text{hyp}} - f^{\text{lin}}|)$ more under hyperbolic funding. Since $|X| \leq \bar{X}_{\text{ret}}$ and the additional funding cost is $|X| \cdot |f^{\text{hyp}} - f^{\text{lin}}| \cdot (\text{time with large } \Delta)$, and the large- Δ period is shorter under hyperbolic (faster rebalancing), the integrated cost difference is bounded by $\bar{s} \cdot \bar{X}_{\text{ret}} \cdot T$. $\square$

The key welfare insight is that hyperbolic funding **increases rewards for liquidity provision during stress** (benefiting arbitrageurs) while **not significantly penalizing insensitive participants** (retail), because the high-funding regime resolves faster.

10 Design Implications and Parameter Selection

10.1 Parameter Selection Framework

The three parameters (R, k, C) control distinct aspects of the mechanism:

1. **Scale R :** Controls the overall magnitude of funding. Higher R produces faster rebalancing but larger peak funding rates. Designers should set R to target a desired near-equilibrium funding sensitivity $\alpha = R(k - 1)/(kC)$.
2. **Curvature k :** Controls where the non-linearity “kicks in.” Lower k (closer to 1) produces earlier non-linearity with a closer asymptote; higher k delays non-linearity but produces a sharper transition. The simulations presented in Section 7 suggest $k \in [1.5, 3.0]$ as the practical range, with $k = 2$ as a default.
3. **Capacity C :** Sets the asymptotic boundary at kC . Should be calibrated to the market’s maximum sustainable imbalance. Specifically, C should be set so that kC exceeds the 99th percentile of historical $|\Delta|$ by a safety margin.

10.2 Adaptive Controller

I propose making parameters time-varying to respond to changing market conditions:

$$\begin{aligned} R_t &= R_0 \cdot \frac{\sigma_t}{\bar{\sigma}} && \text{(volatility-adjusted scale),} \\ C_t &= C_0 \cdot \frac{\text{OI}_{\max} - \text{OI}_t}{\text{OI}_{\max}} && \text{(utilization-adjusted capacity),} \\ k_t &= k_0 + \kappa \cdot \mathbf{1}[\sigma_t > 2\bar{\sigma}] && \text{(regime-switching curvature),} \end{aligned} \tag{27}$$

where σ_t is rolling realized volatility, $\bar{\sigma}$ is its long-run mean, $\text{OI}_{\max}$ is the maximum sustainable OI, and $\kappa > 0$ is a curvature boost during detected stress.

The adaptive controller addresses the oscillation risk identified in Theorem 6.3: by increasing C_t (widening the asymptote) when the market is well-utilized, and decreasing k_t (smoothing the curvature) during volatile periods, the effective response rate remains within the stability bound.

10.3 Connection to Optimal Control

The mechanism design problem can be framed as an optimal control problem:

$$\min_f \int_0^\infty e^{-\rho t} [\Delta_t^2 + \lambda f(\Delta_t)^2] dt \quad \text{s.t.} \quad \frac{d\Delta}{dt} = -\eta f(\Delta) + \xi_t, \tag{28}$$

where the first term penalizes imbalance and the second penalizes excessive funding (UX cost). The parameter λ captures the tradeoff between rebalancing speed and funding rate smoothness:

- When λ is large (high UX priority): the optimal f is approximately linear.
- When λ is small with a capacity constraint $|\Delta| < \Delta_{\max}$: the optimal f approaches the hyperbolic form.

This provides a principled interpretation of the (R, k, C) parameters in terms of the designer’s objectives.

11 Limitations and Failure Modes

1. **Oscillation near asymptote:** As established in Theorem 6.3, discrete-time implementation can produce oscillatory behavior when imbalance is close to kC . Mitigation: exponential moving average smoothing of the imbalance signal, or the adaptive controller (27).
2. **Strategic behavior near asymptote:** Sophisticated traders may exploit the predictable singularity by front-running expected funding spikes. This is a form of funding rate arbitrage that could exacerbate short-term volatility.
3. **Parameter governance:** The mechanism has three parameters vs. one for linear, increasing governance complexity. Miscalibrated C can lead to either excessive funding (too small) or ineffective non-linearity (too large). On-chain governance of (R, k, C) introduces additional attack surfaces.
4. **Composability:** In DeFi environments where multiple protocols interact, the non-linear funding behavior may create unexpected interactions with liquidation engines, oracle systems, or cross-protocol arbitrage strategies.
5. **Empirical limitations:** The empirical analysis is observational and cannot establish causality. Venue heterogeneity beyond the funding mechanism may confound the results.
6. **Model assumptions:** The MFG analysis assumes a continuum of traders and myopic optimization. In practice, traders are finite and forward-looking, and large traders can strategically influence Δ .

12 Conclusion

I have introduced and analyzed hyperbolic funding rate mechanisms as a principled non-linear alternative to linear funding in perpetual futures markets. The main contributions are:

1. An axiomatic derivation showing the hyperbolic form is the unique simple-pole mechanism satisfying linearity at equilibrium, capacity awareness, monotonicity, and convexity.
2. An extended comparison against double-pole, log-barrier, and piecewise mechanisms, establishing a formal speed–stability tradeoff (Proposition 4.4) and demonstrating that hyperbolic occupies a favorable position on the Pareto frontier.
3. Mean-field equilibrium existence and uniqueness, with Lyapunov stability guarantees and explicit oscillation bounds.
4. Simulation evidence of 10–70% faster rebalancing under severe shocks, with the speedup scaling with shock severity.

5. Welfare analysis showing increased arbitrageur surplus without significant retail harm.

The fundamental insight is that a well-designed funding mechanism should be *capacity-aware*: gentle when the market is balanced, aggressive when it approaches dangerous imbalances. The hyperbolic form achieves this with a natural mathematical structure that admits clean theoretical analysis.

Future work includes: (i) empirical validation using on-chain data from production deployments implementing non-linear funding; (ii) formal analysis of the adaptive controller (27) including convergence guarantees; (iii) extension to multi-asset perpetual markets with correlated funding; (iv) mechanism design under adverse selection (informed vs. uninformed traders); and (v) randomized controlled trials comparing funding mechanisms within a single protocol.

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A Proofs of Auxiliary Results

A.1 Proof of Proposition 3.2

Proof of (i). Taylor expansion of (2) around $\Delta = 0$:

$$f(\Delta) = R \left(\frac{Ck(k-1)}{kC} \cdot \frac{1}{1 - \Delta/(kC)} - (k-1) \right) \quad (29)$$

$$= R \left((k-1) \left(1 + \frac{\Delta}{kC} + O(\Delta^2/C^2) \right) - (k-1) \right) \quad (30)$$

$$= R \cdot \frac{k-1}{kC} \Delta + O(\Delta^2/C^2). \quad (31)$$

Thus $f'(0) = R(k-1)/(kC)$, confirming linearity near equilibrium. $\square$

A.2 Derivation of the Oscillation Bound (Theorem 6.3)

Proof. Consider the discrete map $\Delta_{n+1} = G(\Delta_n)$ where $G(\Delta) = \Delta - \eta\delta t \cdot f(\Delta)$. The fixed point $\Delta^* = 0$ is linearly stable if $|G'(\Delta^*)| < 1$.

At a general point $\bar{\Delta}$:

$$G'(\bar{\Delta}) = 1 - \eta\delta t \cdot f'(\bar{\Delta}). \quad (32)$$

For stability: $-1 < 1 - \eta\delta t \cdot f'(\bar{\Delta}) < 1$. The right inequality gives $\eta\delta t \cdot f'(\bar{\Delta}) > 0$, which holds since all terms are positive. The left inequality gives $\eta\delta t \cdot f'(\bar{\Delta}) < 2$.

Substituting $f'(\Delta) = RCk(k-1)/(kC - \Delta)^2$:

$$\eta\delta t < \frac{2(kC - \bar{\Delta})^2}{RCk(k-1)}. \quad (33)$$

As $\bar{\Delta} \rightarrow kC$, the right-hand side goes to zero, meaning no positive step size maintains stability—the discrete system necessarily oscillates near the asymptote. $\square$

B Simulation Code and Reproducibility

All simulation code is available at [repository URL]. The ABM is implemented in Python using NumPy for computation and Matplotlib for visualization. Key simulation parameters:

Table 2: Default simulation parameters.

Parameter	Symbol	Value	Description
Funding scale	R	0.001	Controls funding magnitude
Curvature	k	2.0	Asymptote at $kC = 1000$
Base capacity	C	500	OI capacity
Response rate	η	150	Effective response to funding
Damping coefficient	λ	0.001	Mean-reversion rate
Noise intensity	σ_ε	3.0	Idiosyncratic noise std. dev.
Momentum agents	n_m	30	Trend followers
Arbitrage agents	n_a	20	Funding arbitrageurs
Retail agents	n_r	40	Bounded rationality
Noise agents	n_n	10	Random trading
MC trials	N	100	Monte Carlo robustness

Random seeds are set explicitly for reproducibility. All figures can be regenerated by running `python funding_sim_v3.py`.